

Literature Review

2.1 Introduction

The demographic transition fitness paradox refers to the empirical observation that, in contemporary high-income populations, individuals with greater socioeconomic resources tend to produce fewer offspring than those with fewer resources. This pattern directly contradicts the central prediction of evolutionary theory that organisms should convert resources into reproduction, and it was first explicitly framed as a theoretical problem by Vining (1986), who described it as "the central theoretical problem of human sociobiology." Nearly four decades later the paradox remains unresolved, despite a substantial empirical and theoretical literature spanning evolutionary anthropology, behavioural ecology, formal demography, sociology, and economics.

This chapter reviews that literature in four parts. Section 2.2 establishes the empirical phenomenon itself and situates it within the broader history of the demographic transition. Section 2.3 surveys the three principal theoretical frameworks that have been advanced to explain the paradox: cultural transmission models, quantity-quality tradeoff models, and multigenerational fitness or maladaptation models. Section 2.4 examines the more recent literature on the so-called J-curve hypothesis, which questions whether the SES-fertility relationship is uniformly negative or instead non-monotonic at the upper end of the SES distribution. Section 2.5 reviews the smaller body of work on the individual-level mechanisms by which education in particular depresses fertility. Section 2.6 identifies the methodological and substantive gaps that the present dissertation seeks to address.

2.2 The Demographic Transition and the Emergence of the Paradox

The demographic transition is the historical shift, beginning in eighteenth-century France and subsequently spreading across Europe and the rest of the world, from a regime of high fertility and high mortality to one of low fertility and low mortality (Notestein, 1945; Caldwell, 1976). Across pre-transition agrarian populations, the relationship between wealth or status and reproductive success was typically positive: wealthier landholders, higher-ranking nobles, and more successful farmers had more surviving children than their poorer counterparts (Boone, 1986; Volland, 1990; Pettay et al., 2007). Reviews of historical demographic and ethnographic studies suggest that this positive gradient was the dominant pattern across cultures and time periods prior to industrialisation (Hopcroft, 2006; Nettle & Pollet, 2008).

The reversal of this gradient happened unevenly across populations and over an extended historical window. In Sweden, where unusually complete parish records permit detailed multigenerational analysis, the wealth-fertility relationship weakened during the nineteenth century and inverted in the early twentieth (Dribe, Hacker, & Scalone, 2014; Goodman, Koupil, & Lawson, 2012). In Britain, the timing was broadly similar, with negative gradients well established by the interwar period (Coleman & Salt, 1992). In southern and eastern Europe the inversion was later, in some cases not until the second half of the twentieth century. By the mid-twentieth century, the negative gradient between education or income and completed fertility was a defining feature of fertility patterns across the developed world, and it has persisted to the present in most countries despite considerable variation in its magnitude and shape (Skirbekk, 2008; Andersson et al., 2009).

This inversion was not anticipated by classical demographic transition theory, which framed fertility decline primarily as a response to falling child mortality and to the changing economic value of children, without specifically predicting that fertility would fall fastest among those best resourced to support large families (Mason, 1997). The puzzle that the inversion poses for evolutionary theory was articulated most influentially by Vining (1986), who pointed out that no equivalent reversal had been documented in any other species and that the human pattern therefore required a specifically human explanation. This framing has structured the subsequent literature: most attempts to resolve the paradox accept the empirical fact of the gradient and seek to provide a mechanism that reconciles it with broader evolutionary theory.

2.3 Theoretical Explanations

Three principal explanatory frameworks have dominated the literature. They are not mutually exclusive—several authors argue for combinations of mechanisms—but they make different predictions about the conditions under which the gradient should appear, the populations among which it should be steepest, and the long-run trajectory of fertility.

2.3.1 Cultural Transmission

The cultural transmission framework argues that the demographic transition propagates not through individual rational responses to changed economic conditions but through the social learning of low-fertility norms, typically from higher-status to lower-status individuals (Bongaarts & Watkins, 1996; Cleland, 2001). Bongaarts and Watkins's (1996) analysis of fertility decline across 69 countries showed that variation in the timing of decline was poorly predicted by economic indicators alone but well predicted by indicators of social interaction and information flow, lending support to a diffusion interpretation. Subsequent work has formalised this perspective within cultural evolutionary theory (Boyd & Richerson, 1985; Mesoudi, 2011), modelling fertility norms as transmissible cultural traits subject to prestige bias—preferential learning from high-status models—and to conformity bias, the tendency to adopt majority behaviours.

Colleran (2016, 2020) has been particularly influential in arguing that the cultural-evolutionary account explains features of fertility decline that economic accounts struggle with: the rapidity of decline once it begins, the importance of community-level effects beyond individual circumstances, and the persistence of low fertility even where the economic conditions that initially favoured it have changed. Empirical support comes from studies showing that fertility behaviour clusters geographically and within social networks beyond what individual characteristics would predict (Watkins, 1991; Balbo & Barban, 2014), and from agent-based simulations that reproduce historical patterns of decline when calibrated with social-learning parameters (Colleran & Mace, 2015).

The cultural transmission framework, if correct, implies that the SES-fertility gradient reflects the position of higher-SES individuals as cultural innovators or earlier adopters of low-fertility norms rather than their differential economic incentives per se. It predicts that the gradient should be steepest during the inversion phase of the transition and may flatten or reverse once low-fertility norms become near-universal. It also makes a distinctive prediction at the cross-country level: variation in the timing of inversion should be predicted by indicators of cultural and informational transmission—secularisation, mass education, media penetration, urban-rural integration—rather than by economic indicators alone.

2.3.2 Quantity-Quality Tradeoff and Embodied Capital

The quantity-quality framework, originating in Becker's (1960) economic theory of fertility and developed for evolutionary purposes by Kaplan (1996; Kaplan & Lancaster, 2003), argues that modern parents face an unprecedented incentive to substitute investment per child for number of children. Under conditions of high returns to human capital, parents who concentrate resources on fewer children can produce offspring with substantially greater economic—and potentially reproductive—prospects than parents who spread resources thinly across many children. The framework treats fertility as a strategic allocation problem in which the optimal family size depends on the elasticity of child outcomes to parental investment.

The framework is theoretically appealing because it provides a clear evolutionary logic for low fertility: if the marginal fitness return to a smaller number of well-invested offspring exceeds that of a larger number of less-invested offspring, natural selection should favour the smaller-family strategy. Lawson and Mace (2011) have argued for this position using sibling-competition evidence: children in larger families show reduced cognitive, educational, and health outcomes, consistent with resource dilution at the within-family level.

The principal challenge to the framework is empirical. If quantity-quality substitution were operating optimally, we would expect the fertility-investment tradeoff to maximise long-run fitness measured in grandchildren or great-grandchildren rather than children. Goodman, Koupil, and Lawson (2012), using multigenerational Swedish data, found that low fertility increases descendant socioeconomic position but does not maximise long-run reproductive success—on the contrary, larger families produce more grandchildren on average, even after adjusting for the lower per-child outcomes. This finding suggests that the quantity-quality tradeoff, if it is occurring at all, is not actually maximising biological fitness, and therefore that an additional explanation is required for why parents systematically choose the high-investment strategy.

2.3.3 Maladaptation and Evolutionary Lag

A third framework, less developed but recurrently invoked, treats modern low fertility as a maladaptive response to evolutionary novelty. On this view, the psychological mechanisms that historically converted resources into reproductive success—mate preferences, parental investment heuristics, status-seeking—are not well calibrated to modern conditions of effective contraception, extended education, urban living, and welfare provision (Pérusse, 1993; Kanazawa, 2003). Resources are diverted into status competition or consumption rather than reproduction, not because doing so maximises current fitness but because the underlying psychology was shaped by ancestral environments in which status and reproduction were tightly coupled. The framework predicts that fertility behaviour should respond to cues that were reliable in ancestral environments (perceived status, resource security) even when those cues no longer track current reproductive opportunities.

The maladaptation framework has the virtue of making sense of the multigenerational fitness evidence: if the demographic transition simply represents an evolutionary mismatch, there is no reason to expect long-run fitness to be maximised by current fertility behaviour. Its weakness is that it is often invoked descriptively rather than tested rigorously, and that it sits uneasily with evidence that fertility does respond to current economic conditions (Sobotka, Skirbekk, & Philipov, 2011), implying that the underlying psychology is not entirely

insensitive to current circumstances. The framework is best understood as complementary to the cultural and economic accounts rather than a stand-alone alternative.

2.4 The J-Curve and Recent Developments

A more recent literature has questioned the assumption that the SES-fertility relationship is uniformly negative. Several studies have documented that the relationship may be non-monotonic, with fertility ceasing to decline or rising at the very top of the SES distribution. Hopcroft (2006, 2015), using US General Social Survey data, has reported positive fertility-income gradients among married high-income men alongside negative gradients overall, and has interpreted this as evidence that the paradox is partly an artefact of pooling married and unmarried individuals. Fieder and Huber (2007) found positive fertility-status relationships among men in contemporary Swedish data once unmarried childlessness was accounted for. Nettle and Pollet (2008) found similar patterns across several Western populations.

Stulp and Barrett (2016) provide the most systematic review of the J-curve evidence. They conclude that the relationship is heterogeneous across populations, sub-groups, and historical periods, and that simple statements of either a "negative" or a "positive" gradient obscure considerable underlying variation. They argue, importantly, that the question of whether the modern SES-fertility relationship constitutes a paradox depends on which sub-population and which fertility measure is examined: gradients differ between men and women, between married and unmarried individuals, and between completed cohort fertility and period-based measures.

The J-curve has substantial theoretical implications. If fertility revives at the top of the SES distribution, the paradox may be more limited in scope than originally framed—affecting the middle of the distribution but not its extremes. This would imply that the underlying psychology of resource-fertility conversion is partially intact, and that low fertility among middle-SES populations reflects specific features of their economic and cultural environment rather than a general maladaptation. It would also imply different empirical signatures across countries depending on the shape of their SES distribution and the location of the inflection point.

Despite its importance, the J-curve literature remains methodologically uneven. Few studies use education-stratified completed cohort fertility—the most appropriate outcome for assessing lifetime reproductive success—across multiple countries and cohorts, and the use of modern distributional methods such as quantile regression or non-parametric splines is rare. Most published evidence relies on cross-sectional surveys, single-country designs, or period rather than cohort measures, leaving the cross-national robustness of the J-curve substantially unsettled.

2.5 Individual-Level Mechanism Studies

Alongside the aggregate and cross-country literature, a smaller body of work has examined the individual-level mechanisms by which education in particular depresses fertility. Skirbekk (2008) reviews this literature comprehensively, identifying three principal pathways. The enrolment effect operates through the timing of family formation: extended schooling postpones childbearing, contributing to tempo effects and sometimes leading to unintended childlessness when postponement extends past the biologically feasible window. The opportunity cost effect operates through forgone earnings: higher education increases the wage cost of time spent on childbearing and childrearing, providing an economic incentive to limit family size. The preference effect operates through values and attitudes: education changes views on gender roles, individual autonomy, ideal family size, and the relative importance of career and family.

Disentangling these pathways is methodologically demanding because the endogeneity of education makes naïve regressions of fertility on education unreliable—unobserved characteristics may drive both. Studies using quasi-experimental variation in education, particularly compulsory schooling reforms, have found that exogenous increases in education do reduce fertility, but the estimated effects are smaller than the cross-sectional associations and operate primarily on timing rather than on completed fertility (Monstad, Propper, & Salvanes, 2008; Cygan-Rehm & Maeder, 2013). This pattern is consistent with a substantial role for the enrolment effect but suggests that the opportunity cost and preference effects are smaller than naïve estimates imply.

Snopkowski and Sear (2013), using Indonesian data, found that the attitudinal pathway was substantially more important than the economic pathway in predicting fertility intentions among educated women, supporting cultural transmission interpretations. The British cohort studies, particularly NCDS 1958 and BCS70 1970, have been used extensively in adjacent literatures. Berrington (2017) used BCS70 to examine the divergence between fertility intentions and realised fertility, showing that delayed childbearing increasingly resulted in unintended childlessness for highly educated women. Despite this substantial use, formal mediation analyses of the education-fertility relationship using the BCS70 cohort, separating economic from attitudinal pathways with modern causal-inference methods, remain rare.

2.6 Gaps in the Literature and Contribution of the Present Study

Three principal gaps in the existing literature motivate the present dissertation.

First, despite the theoretical importance of the J-curve hypothesis for interpreting the paradox, systematic tests using education-stratified completed cohort fertility data across multiple European countries are lacking. The existing evidence is fragmentary, drawn from different countries with different methodologies, and rarely uses formal distributional methods to characterise the full shape of the gradient. A multi-country, multi-cohort analysis using quantile regression and semi-parametric splines would constitute a meaningful advance on the existing evidence base.

Second, the existing literature on cross-country variation in the SES-fertility gradient tends to test economic and cultural explanations with a small number of pre-selected covariates in small-sample country-level regressions, an approach that risks both omitted-variable bias and overfitting. A principled high-dimensional analysis using regularised machine learning combined with interpretable variable-importance methods (Athey & Wager, 2019; Lundberg & Lee, 2017) followed by formal Bayesian estimation of the effects identified as substantively important would represent a methodological advance and may yield substantive insight into the relative weight of cultural and economic mechanisms.

Third, the individual-level mediation of education's effect on fertility has been examined in fragmented form but has rarely been subjected to formal causal mediation analysis using modern methods for handling high-dimensional confounding (Chernozhukov et al., 2018). The BCS70 cohort, with its rich attitudinal and longitudinal economic measures and recently released Age 51 sweep providing essentially complete fertility outcomes, is well suited to such an analysis but has not yet, to the candidate's knowledge, been used for this purpose.

The present dissertation addresses each of these gaps. Hypothesis 1 establishes the descriptive pattern of the SES-fertility gradient across European cohorts and countries using education-stratified cohort fertility data. Hypothesis 2 tests the J-curve formally using quantile and semi-parametric methods across multiple cohorts. Hypothesis 3 conducts a high-dimensional comparison of cultural and economic explanations of cross-country variation using interpretable machine learning followed by Bayesian hierarchical modelling. Hypothesis 4 conducts an individual-level mediation analysis of the education-fertility relationship using double machine learning on the BCS70 cohort. Together, these analyses are intended to provide a more rigorous and theoretically integrated account of the contemporary SES-fertility gradient than has hitherto been available, and to contribute empirically to the long-running debate over how the demographic transition fitness paradox should be understood and ultimately explained.